Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Certified Kubernetes Administrator (CKA)

TGS Ref No: TGS-2025054612

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	02 July 2026	First version -- 3.5-day lesson plan aligned to 31 CKA labs; Days 1-3: 9:00 AM - 6:00 PM; Day 4: half-day training + mock + assessment	Tertiary Infotech Academy Pte Ltd

TABLE OF CONTENTS

Right-click and choose “Update Field”, or press F9, to build the table of contents.

Course Title	Certified Kubernetes Administrator (CKA)
Course Code	TGS-2025054612
Duration	3.5 Days (28 training hours)
Daily Schedule (Days 1-3)	9:00 AM - 6:00 PM (8 training hours/day, excluding lunch)
Day 4 Schedule	9:00 AM - 12:00 PM training; 12:00-1:00 PM lunch; 1:00-3:30 PM mock exam; 3:30-3:45 PM tea break; 3:45-6:00 PM assessment briefing & final assessment
Lunch Break	1:00 PM - 2:00 PM (Days 1-3) | 12:00 PM - 1:00 PM (Day 4)
Breaks	Tea breaks scheduled within each day
Delivery Mode	Instructor-led, hands-on labs in Killercoda browser terminals
Prerequisites	Linux command-line familiarity; basic Kubernetes awareness helpful
Tools	kubectl, kubeadm, Helm 3, crictl, etcdctl, Killercoda playgrounds, killer.sh

Course Overview

This 3½-day hands-on course prepares participants for the Certified Kubernetes Administrator (CKA) certification examination. Through 31 structured Killercoda labs and instructor-led sessions, participants gain practical skills in cluster installation, configuration, workloads, networking, storage, and troubleshooting — aligning with the five CKA exam domains: Cluster Architecture, Installation & Configuration (25%); Workloads & Scheduling (15%); Services & Networking (20%); Storage (10%); and Troubleshooting (30%).

Learning Outcomes

By the end of this course, participants will be able to:

- Bootstrap a Kubernetes cluster from scratch using kubeadm with a CNI plugin and validate the control plane.
- Upgrade a cluster using kubeadm from one minor version to the next with zero-downtime drain and uncordon.
- Configure high-availability control plane nodes with HAProxy load balancer and Keepalived VIP.
- Manage cluster access with RBAC: create Roles, ClusterRoles, RoleBindings, and ServiceAccounts.
- Package and deploy applications using Helm 3 charts; manage repos, upgrades, and rollbacks.
- Apply Kustomize overlays for environment-specific configuration without templating.
- Extend Kubernetes with Custom Resource Definitions (CRDs) and deploy Operators.

- Manage Deployment rollouts, rolling updates, and rollbacks; configure HPA v2 for autoscaling.
- Inject configuration and secrets into Pods via ConfigMaps and Secrets (env and volume mounts).
- Configure Pod scheduling using nodeSelector, node affinity, taints, tolerations, and PriorityClasses.
- Expose services using ClusterIP, NodePort, Ingress controllers, and Gateway API with TLS.
- Isolate network traffic with NetworkPolicy rules and troubleshoot DNS with CoreDNS.
- Provision persistent storage with PersistentVolumes, PersistentVolumeClaims, and StorageClasses.
- Diagnose and restore failing cluster components, nodes, and application Pods.
- Monitor cluster health with kubectl top, events, and metrics; troubleshoot networking and RBAC failures.

Day 1 — Cluster Architecture, Installation & Configuration

Domain 1 — Cluster Architecture, Installation & Configuration (CKA 25%)

Morning Session — 9:00 AM – 1:30 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:15	Registration & Setup — Killercoda, kubectl aliases, kubeconfig verify	15 min	2–11
9:15 – 9:45	Welcome & CKA Exam Overview — domains, weights, exam environment	30 min	12–22
9:45 – 10:30	Lab 1: kubeadm Prerequisites — kernel modules, sysctls, containerd, kubeadm/kubelet/kubectl install	45 min	23–26
10:30 – 11:30	Lab 2: kubeadm Bootstrap — init, kubeconfig, PKI, join worker	60 min	27–31
11:30 – 11:45	Tea Break	15 min	—
11:45 – 12:30	Lab 3: Install Calico CNI — tigera-operator, node Ready, pod network test	45 min	33–40
12:30 – 1:30	Lab 4: Cluster Upgrade — v1.34 to v1.35, drain/uncordon, verify	60 min	41–48

Time	Topic / Activity	Duration	Slides
1:30 PM – 2:30 PM	Lunch Break	60 min	49

Afternoon Session — 2:30 PM – 6:00 PM

Time	Topic / Activity	Duratio	Slide

		n	s
2:30 – 3:30	Lab 5: HA Control Plane — HAProxy, Keepalived VIP, second control-plane join	60 min	50–59
3:30 – 4:15	Lab 6: Helm — repos, install, upgrade, rollback, uninstall	45 min	60–68
4:15 – 4:30	Tea Break	15 min	—
4:30 – 5:15	Lab 7: Kustomize — base/overlays, namePrefix, image transformer	45 min	69–76
5:15 – 6:00	Lab 8: RBAC — Role, RoleBinding, ServiceAccount, ClusterRole, auth can-i	45 min	78–85
6:00	Day 1 Review & Q&A	—	86

Day 2 — Workloads & Scheduling + Services & Networking

Domain 2 — Workloads & Scheduling (CKA 15%) + Domain 3 — Services & Networking (CKA 20%)

Morning Session — 9:00 AM – 12:30 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:15	Day 2 Intro — review Day 1, CKA Workloads domain overview	15 min	88–90
9:15 – 10:00	Lab 9: CRDs & Operators — AppDatabase CRD, schema validation, Operator deploy	45 min	91–99
10:00 – 10:45	Lab 10: Extension Interfaces — crictl, /etc/cni/net.d, CSI driver inspect	45 min	100–103
10:45 – 11:00	Tea Break	15 min	—
11:00 – 11:45	Lab 11: Deployments & Rollouts — rolling update, rollback, maxSurge/Unavailable	45 min	105–114
11:45 – 12:30	Lab 12: ConfigMaps — 3 creation methods, envFrom, volumeMount	45 min	116–124

Time	Topic / Activity	Duration	Slides
12:30 PM – 1:30 PM	Lunch Break	60 min	115

Afternoon Session — 1:30 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
1:30 – 2:15	Lab 13: Secrets — generic/TLS/registry, tmpfs mount, env inject	45 min	125–128
2:15 – 3:00	Lab 14: HPA v2 — metrics-server, autoscale, load test, observe	45 min	130–138
3:00 – 3:15	Tea Break	15 min	—
3:15 – 4:00	Lab 15: Self-Healing — liveness/readiness probes, restartPolicy	45 min	139–144
4:00 – 4:45	Lab 16: Scheduling — nodeSelector, node affinity, taints/tolerations, PriorityClass	45 min	145–153

4:45 – 5:30	Lab 17: Pod Connectivity — flat network, ping, curl, DNS resolution	45 min	154–160
5:30 – 6:00	Lab 18: Service Types — ClusterIP, NodePort, MetalLB LoadBalancer	30 min	161–164
6:00	Day 2 Review & Q&A	—	165

Day 3 — Services & Networking + Storage + Troubleshooting

Domain 3 — Services & Networking (cont.) + Domain 4 — Storage (CKA 10%) + Domain 5 — Troubleshooting (CKA 30%)

Morning Session — 9:00 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:15	Day 3 Intro — Networking & Storage domain overview	15 min	167–169
9:15 – 10:00	Lab 19: Ingress Controller — host routing, TLS termination, path rules	45 min	170–180
10:00 – 10:45	Lab 20: Gateway API — GatewayClass, Gateway, HTTPRoute, traffic split	45 min	181–186
10:45 – 11:00	Tea Break	15 min	—
11:00 – 11:45	Lab 21: NetworkPolicy — default-deny, pod selector, namespace selector	45 min	188–199
11:45 – 12:15	Lab 22: CoreDNS — Corefile, headless services, stubDomain	30 min	200–203
12:15 – 1:00	Lab 23: PV & PVC — static provisioning, access modes, reclaim policy	45 min	205–216

Time	Topic / Activity	Duration	Slides
1:00 PM – 2:00 PM	Lunch Break	60 min	204

Afternoon Session — 2:00 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:45	Lab 24: StorageClass — dynamic provisioning, default class, ReadWriteMany	45 min	217–222
2:45 – 3:30	Lab 25: Volume Types — emptyDir, hostPath, projected volumes	45 min	223–227
3:30 – 3:45	Tea Break	15 min	—
3:45 – 4:45	Lab 26: Troubleshoot Cluster Components — API server, etcd, scheduler faults	60 min	229–236
4:45 – 5:45	Lab 27: Troubleshoot Nodes — NotReady, kubelet failures, node repair	60 min	237–240
5:45 – 6:00	Lab 28: Application Logs — kubectl logs, --previous, multi-container	15 min	241–247
6:00	Day 3 Review & Q&A	—	248

Day 4 (Half Day) — Troubleshooting + Mock Exam + Assessment

Domain 5 — Troubleshooting (cont.) + Final Assessment from 4:00 PM

Morning Training — 9:00 AM – 12:00 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:45	Lab 29: Monitor Cluster — kubectl top, events, metrics-server, resource usage	45 min	250–260
9:45 – 10:30	Lab 30: Troubleshoot Networking — DNS, service, endpoint, kube-proxy	45 min	262–270
10:30 – 10:45	Tea Break	15 min	—
10:45 – 11:30	Lab 31: Troubleshoot RBAC & Scheduling — 403 Forbidden, Pending pods	45 min	271–278
11:30 – 12:00	CKA Exam Tips, Time Management, killer.sh walkthrough	30 min	279

Time	Topic / Activity	Duration	Slides
12:00 PM – 1:00 PM	Lunch Break	60 min	280

Afternoon — 1:00 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
1:00 – 2:30	Mock Exam Practice (killer.sh environment or timed kubectl tasks)	90 min	281–282
2:30 – 3:30	Mock Exam Debrief & Common Mistakes Review	60 min	—
3:30 – 3:45	Tea Break	15 min	—
3:45 – 4:00	Assessment Briefing — WSQ assessment instructions and rules	15 min	284
4:00 – 6:00	WSQ Assessment — practical CKA-style assessment	120 min	285–286
6:00	Assessment submission & close	—	287

Labs Covered

All 31 hands-on labs are delivered in Killercoda browser terminals and mirror the slide decks and Learner Guide exactly:

Lab	Domain	Topic	Title
1	Cluster Architecture, Installation & Configuration	kubeadm setup	kubeadm Prerequisites & Container Runtime

2	Cluster Architecture, Installation & Configuration	kubeadm bootstrap	Bootstrap Control Plane with kubeadm init
3	Cluster Architecture, Installation & Configuration	CNI	Install Calico CNI
4	Cluster Architecture, Installation & Configuration	Cluster upgrade	Cluster Upgrade with kubeadm
5	Cluster Architecture, Installation & Configuration	High availability	HA Control Plane with HAProxy & Keepalived
6	Cluster Architecture, Installation & Configuration	Helm	Helm Package Manager
7	Cluster Architecture, Installation & Configuration	Kustomize	Kustomize Overlays
8	Cluster Architecture, Installation & Configuration	RBAC	Role-Based Access Control
9	Workloads & Scheduling	CRDs & Operators	Custom Resource Definitions & Operators
10	Workloads & Scheduling	Extension interfaces	CRI, CNI, CSI Extension Interfaces
11	Workloads & Scheduling	Deployments	Deployments and Rollouts
12	Workloads & Scheduling	ConfigMaps	ConfigMaps – Env & Volume Injection
13	Workloads & Scheduling	Secrets	Secrets Management
14	Workloads & Scheduling	HPA	Horizontal Pod Autoscaler v2
15	Workloads & Scheduling	Self-healing	Self-Healing with Probes & Restart Policies
16	Workloads & Scheduling	Scheduling	Pod Scheduling – Affinity, Taints & Tolerations
17	Services & Networking	Pod networking	Pod Connectivity and Flat Network
18	Services & Networking	Service types	ClusterIP, NodePort, and MetalLB LoadBalancer
19	Services & Networking	Ingress	Ingress Controller with Host Routing & TLS
20	Services & Networking	Gateway API	Gateway API – GatewayClass, Gateway, HTTPRoute
21	Services & Networking	NetworkPolicy	Network Policies – Default-Deny and Allow Rules
22	Services & Networking	CoreDNS	CoreDNS – Service Discovery and DNS Resolution
23	Storage	PV & PVC	PersistentVolumes and PersistentVolumeClaims
24	Storage	StorageClass	Dynamic Provisioning with StorageClass
25	Storage	Volume types	emptyDir, hostPath, and Projected Volumes
26	Troubleshooting	Cluster components	Troubleshoot Cluster Components
27	Troubleshooting	Node troubleshooting	Troubleshoot Nodes – NotReady & kubelet
28	Troubleshooting	Logs & debugging	Application Logs and Pod Debugging
29	Troubleshooting	Monitoring	Monitor Cluster with kubectl top & Metrics Server
30	Troubleshooting	Network troubleshooting	Troubleshoot Networking – DNS, Services, Endpoints
31	Troubleshooting	RBAC & scheduling	Troubleshoot RBAC Errors and Scheduling Failures

Assessment

Participants are assessed through their hands-on lab work across Days 1–4 and a final written / MCQ assessment on Day 4 afternoon. The assessment tests practical CKA-domain skills including cluster administration, workload management, networking, storage, and troubleshooting. A minimum competency score must be achieved to satisfy WSQ requirements.